Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/23/2019
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/22/2016	**End Date:**	11/21/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement’s material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments will satisfy two major objectives. First, the microstructure and material properties of the benchmark samples will be evaluated; different responses to thermal and mechanical loading are expected and will be characterized in detail. Any positive attributes found in the special environment will be reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or usage can have huge implications on the sustainability of the cement industry.
Task Progress:	Since the return of the samples from the ISS on August 13th of 2018, extensive microscopic investigations were carried out at Marshall Space Flight Center (MSFC) and Penn State University. 10 out of 16 series tested were imaged and the most pronounced effects of µg on cement hydration were identified. Subsequently, a few series, including Series 1, 2, 5, 8, 9, and 15, were selected for a more in-depth analysis and comparison to the ground control samples. In addition to the cement pastes, an expansion on the raw lunar regolith simulant (Johnson Space Center (JSC) 1A)) research was carried out, which provided better understanding of the material`s class and reactivity. Task 1: Analysis of 1g and µg cement samples. Several tests were performed to evaluate the effect of µg on cement hydration. The observed microstructural changes are attributed to the more uniform distribution of gaseous, liquid, and solid phases. The uniform phase distribution takes place at very early stages of hydration (i.e., when the paste is still in fluid state) due to lack of buoyancy and sedimentation. Task 2: Supplementary ground research. 2.1. Clinostat: A clinostat is theoretically able to mimic microgravity conditions by rotating a sample at a slow speed in the horizontal axis to average out the forces of gravity. Mixtures were mounted on the tip of a needle and suspended in water to try and match results seen within the microgravity samples of Series 1 and 2. The results of the experiment were mixed and did not always necessarily match the actual microgravity samples. 2.2. Alcohol Study: The samples that were flushed with alcohol during original mixing have been sitting in the alcohol since the initial mix. There is no knowledge on whether prolonged alcohol exposure may influence the samples. As a start, anhydrous C¬3A and gypsum were placed in alcohol and the morphology is being analyzed via SEM at various ages. So far, no images have revealed any signs to be concerned with the prolonged exposure. Task 3: Lunar Regolith simulant study (continued). The reactivity of raw JSC 1A was tested and its amorphous content determined. This was done to further characterize the regolith simulant as a construction material and better interpret the characteristics of JSC-Portland cement blended mixtures, which was explained in details in the last annual report. Additional tests were conducted as follows: 3.1. Quantitative XRD: The test was used to quantify the amorphous content and each of crystalline phases composing JSC 1A. Although several studies were conducted to characterize JSC 1A, this piece of information was not found in the literature. The test was performed with a Cobalt x-ray source to ensure accurate measurements of the Iron-containing phases. The results show that 66% (by mass) of the regolith simulant is amorphous, which potentially contribute to its reactivity. 3.2. Reactivity test: The reactivity of three particle size ranges of JSC-1A was directly assessed by performing a test that evaluates the heat release (using isothermal calorimetry) and subsequently CH consumption (using TGA). Once the heat release and CH consumption are determined, JSC-1A can then be classified into inert, pozzolanic, or latent hydraulic material. Despite the high amorphous content (66%), the raw JSC-1A (i.e., full gradation) appears to be inert, whereas the portion finer than 75 µm was classified as a less reactive pozzolan.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Abstracts for Journals and Proceedings	Neves J, Scheetz B, Grugel R, Radlinska A. "Experimental investigation of cement hydration in gravity-free environment." 15th International Congress on the Chemistry of Cement, Prague, Czech Republic, September 16-20, 2019. Conference proceedings. 15th International Congress on the Chemistry of Cement, Prague, Czech Republic, September 16-20, 2019. , Sep-2019
Articles in Peer-reviewed Journals	Neves J, Collins P, Wilkerson R, Grugel R, Radlinska A. "Microgravity effect on microstructural development of tri calcium silicate (C3S) paste." Frontiers in Materials. 2019 Apr. Published on 24 April 2019. https://doi.org/10.3389/fmats.2019.00083 , Apr-2019